

Problem Set 1

INTRODUCTION

Problem 1 Aggregate Output. During a given year, the following activities occur:

- A silver mining company pays its workers \$200,000 to mine 75 pounds of silver. The silver is then sold to a jewelry manufacturer for \$300,000.
 - The jewelry manufacturer pays its workers \$250,000 to make silver necklace, which the manufacturer sells directly to consumers for \$1,000,000.
- 1) Using the production-of-final goods approach, what is GDP in this economy?
 - 2) What is the value added at each stage of production? Using the value-added approach, what is GDP?
 - 3) What are the total wages and profits earned? Using the income approach, what is GDP?

You may find it helpful to fill in the table below:

Sliver Mining Company		Jewelry Manufacturer	
Revenues from sales	\$	Revenues from sales	\$
Expenses	\$	Expenses	\$
Wages	\$	Wages	\$
		Silver purchases	\$
Profit	\$	Profit	\$

THE GOODS MARKET

Problem 1 Balanced Budget Multiplier. Consider the equilibrium condition in the goods market:

$$Y^* = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T].$$

- 1) Suppose the government decides to increase T and G by one unit at the same time. What is the change in equilibrium output?
- 2) Based on you answer to 1), comment on the idea that “fiscal policy is macroeconomically neutral if balanced budget is maintained”.
- 3) How will the change in the propensity to consumer affect your answer to 1)? Why?

Problem 2 Automatic Stabilizer. *In the lecture we assume the fiscal policy to be exogenous. In reality, however, taxes typically depend on the level of the income and tend to be higher when income is higher. In this question we consider a case when this automatic response of taxes can help “stabilize” the economy when there is a change to autonomous spending.*

Consider the following equations:

$$C = c_0 + c_1(Y - T)$$

$$T = t_0 + t_1 Y$$

$$Y = C + \bar{I} + G.$$

- 1) Solve for equilibrium output.
- 2) What is the multiplier? How does the change of t_1 affect the multiplier? Explain your answer.
- 3) Why is fiscal policy in this case called automatic stabilizer?

Problem 3 Endogenous Investment. *This problem examines the implications of allowing investment to depend on output. IS-LM model carries this analysis much further and introduces an essential relation—the effect of the interest rate on investment—not examined in this problem.*

Suppose the economy is characterized by:

$$C = c_0 + c_1(Y - T)$$

$$I = b_0 + b_1Y$$

$$Y = C + I + G.$$

- 1) Solve for equilibrium output.
- 2) What is the value of the multiplier? How does the relation between investment and output affect the value of the multiplier? For the multiplier to be positive, what condition must $(c_1 + b_1)$ satisfy? Explain your answers.
- 3) What would happen if $(c_1 + b_1) > 1$?
- 4) Suppose that the parameter b_0 , sometimes called *business confidence*, increases. How will equilibrium output be affected? Will investment change by more or less than the change in b_0 ? Why? What will happen to national saving?

THE FINANCIAL MARKET

Problem 1 Money Demand vs. Bonds Demand. Suppose that a person's wealth is \$50,000 and that her yearly income is \$60,000. Also suppose that her money demand function is given by

$$M^d = \$Y(0.35 - i)$$

- 1) Derive the demand for bonds, which is denoted as B^d . Suppose the interest rate increases by 10 percentage points. What is the effect on her demand for bonds? (*Hint: Demand for money and demand for bonds must add up to her wealth.*)
- 2) What are the effects of an increase in wealth on her demand for money and her demand for bonds? Explain in words.
- 3) What are the effects of an increase in income on her demand for money and her demand for bonds? Explain in words.

Problem 2 Money and Banking System. In this question, let us assume that

- The public holds no currency;
- The ratio of reserves to deposits is 0.1, and no bank holds excessive reserve;
- The demand for money is given by

$$M^d = \$Y(0.8 - 4i).$$

Initially, the supply of central bank money is \$100 billion, and nominal income is \$5 trillion.

- 1) What is the demand for central bank money H^d ? (*Hint: How is it related to M^d , when the public holds no currency and no bank holds excessive reserve?*)

- 2) Find the equilibrium interest rate by setting the demand for central bank money equal to the supply of central bank money.
- 3) What is the overall supply of money? Is it equal to the overall demand for money at the interest rate you found in 2)?
- 4) What is the effect on the interest rate if central bank money is increased to \$300 billion?
- 5) If the overall money supply increases to \$3,000 billion, what will be the effect on i ?

THE $IS - LM$ MODEL

Problem 1 Numerical IS-LM. Consider the following numerical example of the $IS - LM$ model:

$$C = 200 + 0.25Y_D$$

$$I = 150 + 0.25Y - 1000i$$

$$G = 250$$

$$T = 200$$

$$\bar{i} = 0.05$$

- 1) Derive the IS relation.
- 2) The central bank sets an interest rate of 5%. How is that decision represented in the equations?
- 3) What is the level of real money supply when the interest rate is 5%? Use the expression:

$$\frac{M^d}{P} = 2Y - 8000i$$

- 4) Solve for the equilibrium values of C , and I , and verify the value you obtained for Y by adding C , I , and G .
- 5) Now suppose that government spending increases to $G = 400$. Explain the effects of an expansionary fiscal policy on Y , I , C , and real money supply.
- 6) Starting from an interest rate 5% and government spending equal to 250 units, increase government spending to 400 units while fixing the real money supply at 1600 units. Compare the effect of the increase in the government spending on Y , I , and C in 5) and explain the difference.

Now return to the initial condition in which government spending $G = 250$. Suppose that the central bank cuts the interest rate to 3%.

- 7) How does this change the LM curve? Solve for the equilibrium values for Y , I , C , and real monetary supply. Explain in words the effects of an expansionary monetary policy.

THE FINANCIAL MARKET II

Problem 1 The Troubled Asset Relief Program (TARP). Consider a bank that has assets of 100, capital of 20, and short term credit of 80. Among the bank's assets are securitized assets whose value depends on the price of houses. These assets have a value of 50.

- 1) Set up the bank's balance sheet.

Suppose that as a result of a housing price decline, the value of the bank's securitized assets falls by an uncertain amount, so that these assets are now worth somewhere between 25 and 45. Call the securitized assets "troubled assets." The value of the other assets remains at 50. As a result of the uncertainty about the value of the bank's assets, lenders are reluctant to provide any short-term credit to the bank.

- 2) Given the uncertainty about the value of the bank's assets, what is the range in the value of the bank's capital?

As a response to this problem, the government considers purchasing the troubled assets, with the intention of reselling them again when the markets stabilize. (This is the original version of the TARP.)

- 3) If the government pays 25 for the troubled assets, what will be the value of the bank's capital? How much would the government have to pay for the troubled assets to ensure that the bank's capital does not have a negative value? If the government pays 45 for the troubled assets, but the true value turns out to be much lower, who bears the cost of this mistaken valuation? Explain.

Suppose instead of buying the troubled assets, the government provides capital to the bank by buying ownership shares, with the intention of reselling the shares when the markets stabilize. (This is what the TARP ultimately became.) The government exchanges treasury bonds (which become assets for the bank) for ownership shares.

- 4) Suppose the government exchanges 25 of Treasury bonds for ownership shares. Assuming the worst-case scenario (so that the troubled assets are worth only 25), set up the new balance sheet of the bank. (Remember that the firm now has three assets: 50 of untroubled assets, 25 of troubled assets, and 25 of Treasury bonds.) What is the total value of the bank's capital? Will the bank be insolvent?
- 5) Given your answers and the material in the text, why might recapitalization be a better policy than buying the troubled assets?

THE MEDIUM RUN

Problem 1 The Natural Rate of Unemployment. Suppose that the markup of goods prices over marginal cost is 5%, and that the wage-setting equation is

$$W = P(1 - u)$$

where u is the unemployment rate.

- 1) What is the real wage, as determined by the price-setting equation?
- 2) Solve for the natural rate of unemployment.
- 3) What happens to the natural rate of unemployment if the markup of prices over costs increases to 10%? Explain the logic behind your answer.

Problem 2 Mutations of the Phillips Curve. Suppose that the Phillips curve is given by

$$\pi_t = \pi_t^e + 0.1 - 2u_t$$

and expected inflation is given by

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

and suppose that θ is initially equal to 0 and $\bar{\pi}$ is given and does not change. It could be zero or any positive value. Suppose that the rate of unemployment is initially equal to the natural rate. In year t , the government decides to bring the unemployment rate down to 3% and hold it there forever.

- 1) Determine the rate of inflation in periods t , $t + 1$, $t + 2$, $t + 3$, $t + 4$, $t + 5$.
- 2) Do you believe the answer given in 1)? Why or why not? (Hint: Think about how people are more likely to form expectations of inflation.)

Now suppose that in year $t + 6$, θ increases from 0 to 1. Suppose the government still decides to keep the unemployment at 3% forever.

- 3) Why might θ increase in this way?
- 4) Calculate the inflation rate in years $t + 6$, $t + 7$, and $t + 8$.
- 5) What happens to inflation when $\theta = 1$ and unemployment is kept below the natural rate of unemployment?
- 6) What happens to inflation when $\theta = 1$ and unemployment is kept at the natural rate of unemployment?

THE LONG RUN

Problem 1 The Cobb-Douglas Production Function and the Steady State. Suppose that the economy's production function is given by

$$Y = K^{1/3}N^{2/3}$$

- 1) Transform the production function into a relation between output per worker and capital per worker.
- 2) For a given saving rate s , and depreciation rate δ , derive an expression for capital per worker and output per worker in the steady state.
- 3) Solve for the steady-state level of output per worker when $s = 0.32$ and $\delta = 0.08$.
- 4) Briefly explain about the golden level of consumption. Suppose that the depreciation rate remains constant at $\delta = 0.08$. Solve for the golden-rule saving rate, golden-rule level of capital per worker, golden rule level of output per worker, and golden-rule level of consumption.
- 5) Suppose that the depreciation rate remains constant at $\delta = 0.08$, while the saving rate is reduced to $s = 0.16$. What is the new steady-state output per worker?

Problem 2 Growth Accounting. Assume the production function is given by

$$Y = K^{1/3}(AN)^{2/3}$$

- 1) Show that the rate of growth of total factor productivity $g_A = \frac{3}{2}(g_Y - g_N) - (g_K - g_N)$ where g_Y denotes the growth rate of Y , g_K the growth rate of capital, and g_N growth rate of the labor input.
- 2) What do the quantity $g_Y - g_N$ and $g_K - g_N$ represent, respectively?